Flood Extent Mapping From SAR Images by a Dynamic Threshold: A Case Study of 2022 Compound Flood in Northeastern Bangladesh

Raihan Uddin Ahmed, Md Mamunur Rashid, and Emad H. Habib

***Abstract*—Satellite images have gained considerable traction for near-real-time and rapid flood mapping because they are agnostic to flooding mechanisms and their drivers, hence applicable to any floods, whether pluvial, fluvial, or both (i.e., compound flood). However, flood mapping using satellite images is challenging due to cloud cover and shadows, particularly in regions where floods typically occur during monsoon seasons with frequent and extensive cloud cover (e.g., South Asian countries). Generally, optical images (e.g., Landsat) are used alone or in combination with synthetic aperture radar (SAR) images (e.g., Sentinel-1) under the multisource flood mapping (MSFM) framework for rapid flood detection. However, clouds often affect optical images, limiting their ability to provide reliable flooding information. In addition, the MSFM requires diligent (and often manual) identification of a preflood image, which may be challenging when automatically delineating multiple floods. Moreover, using a unique permanent water body for flood delineation may be erroneous for regions, such as northeastern Bangladesh, where the water extent climatology varies significantly for monsoon months (when floods usually occur). In contrast, this work presents a rapid, simple, and comparable thresholding algorithm for flood delineation using Sentinel-1 SAR images, where backscatter coefficient values are used to identify the watered pixels, and the flooded pixels are determined by comparing them with the monthly climatology of water extent. The method is applied to map the surface water and flood extents for catastrophic back-to-back flood events in the northeastern part of Bangladesh in 2022 caused by heavy local precipitation and upstream discharges (i.e., compound flood), leveraging the cloud-based computing capabilities of Google Earth Engine. The results (i.e., surface water and flood extent maps) are comparable with those obtained using the recently developed sophisticated MSFM and existing SAR-based methods. Besides, our thresholding algorithm provides reliable flood maps, for instance, where MSFM miscalculates flood extent due to significant cloud cover. The technique developed in this study will enable decision makers and emergency responders to obtain precise information rapidly during time-sensitive flooding events, supporting informed decision making for resource allocation and rescue planning.**

***Index Terms*—Compound flooding, floods, Google Earth Engine (GEE), satellite images, Sentinel-1.**

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I. Introduction

FLOODS are among the most recurrent and catastrophic natural disasters [1], [2]. According to the 2017 report from the International Disaster Database, around 85 million people worldwide were affected by floods, which caused an average annual damage of 36.7 billion USD [3]. Floods are driven by various hydrometeorological variables, often called drivers, such as precipitation [4], [5], river discharge [6], [7], and storm surge correspond to pluvial, fluvial, and coastal flooding. However, multiple drivers may work in tandem to generate floods. Compound flooding has recently been conceptualized as a phenomenon where multiple flood-generating drivers (e.g., soil moisture, rainfall, upstream discharge, or two flooding events in close succession) co-occur or nearly co-occur, which may exacerbate the flooding intensity and devastation. For example, localized heavy precipitation can be combined with upstream discharges under a dependence structure to exacerbate the flooding hazard [8], [9]. A few examples of such fluvial–pluvial compound floods are the 2015 Memorial Day flood in Austin, Texas [10], the 2020 Chongqing flood in Southwest China [11], the 2020 flood in the Yangtze River and the Yellow River basins in China [11], [12] and the 2022 flood in the northeastern region of Bangladesh. Until recently, floods have been investigated and analyzed separately, considering each type of flood depending on its drivers, i.e., univariate flood analysis. Recent flood risk analysis and assessment studies consider co-occurrences of different flood-generating drivers due to their underlying dependence [13], [14], [15], [16], [17], [18]. However, they mainly focused on hazard assessment (e.g., joint return periods of flood). Mapping the extent and depth of compound flooding remains relatively underrepresented in the literature due to the computational burden of numerical hydrodynamic models and the challenges in coupling them to integrate pluvial, fluvial, and coastal (storm surge) flooding. Remote sensing data have the potential for mapping compound floods because they are agnostic to flooding mechanisms and drivers, and are applicable to any flood regime, whether pluvial, fluvial, or compound.

Flood mapping is crucial for assessing flood risks, planning mitigation strategies, and setting insurance rates. Earlier studies have modeled flood extent using various approaches [19],

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including satellite images [19], [20], hydrodynamic modeling [21], [22], and machine learning and deep learning techniques [23], [24]. Hydrodynamic models are computationally demanding [25], necessitating the development of reduced physics' models to overcome the computational burden [26], [27]. Typically, these models only accounted for fluvial floods, whereas recent studies have begun to incorporate pluvial floods into state-of-the-art models [26], [28], [29], [30]. The use of satellite images for flood hazard mapping is also gaining popularity due to their preprocessing, public availability from various sources, computational efficiency, and the availability of free cloud-based analysis tools [e.g., Google Earth Engine (GEE)] [31], [32], [33], [34], [35], [36], [37]. Specifically, they are significantly valuable for data-scarce and resource-limited regions, such as Bangladesh. The Global Flood Awareness System (GloFAS) recently added a global flood monitoring (GFM) component that leverages Sentinel-1 satellite images to derive publicly available global flood extent maps [38]. GFM provides reasonable flood information across Bangladesh (critical success index varies from 70% to 80%) [39], [40], and it could be adopted for flood monitoring in countries, such as Bangladesh. However, due to the complexity of the algorithms and global coverage, GFM takes approximately 8 h to provide flood maps after the satellite images are publicly available [39]. While acknowledging the global coverage, huge data integration, and significant computation in producing the GFM data, a simplified algorithm for rapid flood detection with comparable efficiency and provision of executing it in the publicly available cloud-based computation platform (e.g., GEE) within minutes of satellite image availability can be a complementary and considerable option for local real-time flood monitoring because it will save precious time in decision making during catastrophic flood events.

Flood mapping utilizes near-real-time satellite data from Earth observation. Sensor technology and image capture process determine the kind of satellite: optical or synthetic aperture radar (SAR). Optical satellites, such as Landsat, MODIS, and Sentinel-2, capture Earth images using optical sensors. In contrast, SAR satellites (e.g., Sentinel-1) transmit microwaves and use the reflected signal from the surface to build images.

SAR satellites have become popular for flood detection in recent years. Unlike optical satellites, they provide images unaffected by weather conditions, clouds, or the time of day [36], [41], [42]. In SAR images, flooded areas appear darker due to their low backscatter coefficient value, which is the wave reflected toward the radar, compared with other landcover types [43]. Fuzzy logic classification [44], relevance vector machine [45], texture analysis method [46], and change detection algorithms [47] are several methods of flood mapping using SAR data. The threshold-based change detection algorithm for flood detection from SAR images is a popular method for rapid flood mapping [48], [49], [50], [51]. Otsu thresholding method (OTM) is one of the oldest and most popular methods of flood mapping using SAR images [52], [53], [54], [55]. However, the method strongly decreases in performance once the bimodality starts disappearing [56], [57]. Some of the more recent studies use dual-polarization combinations to map floods without prior assumptions, allowing for easier automation [58], [59]. Studies have also been conducted to improve the method by using localized methods through tiling [60], [61]. However, algorithms developed in some recent studies use optical imagery to aid in determining the threshold for the change detection index of SAR images for successfully extracting the flooded pixels [19], [36]. An example of such a method is known as multisource flood mapping (MSFM), which typically uses optical images to calibrate the SAR change detection index threshold [19], [62], [63], [64]. This method may be challenging because optical images provide very little or unreliable data when they contain significant cloud cover. The presence of clouds, their shadows, and snow in the optical images can disrupt the detection of water pixels [65], [66], [67]. To overcome this limitation, studies have suggested various cloud-removing approaches to remove clouds from optical images [65], [68], [69], [70]; however, it may leave limited usable data in the images, making them insufficient for the accurate flood mapping [71], [72]. In Asian countries, such as Bangladesh, where floods occur during the monsoon season and are often accompanied by frequent and considerable cloud cover, using optical satellite images for flood mapping is quite challenging. In this case, methods deriving flood maps solely from SAR imagery could be an option. Nevertheless, there are a few disadvantages to using only SAR imagery. A nearly perfect dihedral corner reflector forms when floodwaters flood streets and open spaces between structures. The incident SAR signal reflects horizontally from the smooth water surface toward a nearby vertical building wall and then reflects off the wall back toward the sensor, causing an exceptionally strong signal return, causing inundated urban areas to appear as anomalously bright pixels in the SAR image [73], [74]. Furthermore, the strength and detectability of this double-bounce signal are not uniform, further adding to the complexity [75]. Elevated structures obstruct the radar beam from reaching the ground surface located behind them, causing darkened sections to lack signal reception, resulting in black zones with little backscatter readings that are indistinguishable from still water [76]. Large inundated rivers or lakes subjected to intense winds generate capillary waves that induce Bragg scattering, consequently increasing the SAR return and creating erroneous bright patches on the water surface, which compromises threshold-based water extraction [77], [78], [79]. Heavy rainfall can also change the roughness of water, thereby hindering the detection of floods [80]. SAR also has a limited record length.

The record-breaking back-to-back flooding across northeastern Bangladesh in May and June 2022 severely damaged infrastructure, disrupted society, and harmed the environment. They affected 7.2 million people, killed over 68, and infected over 4000 with waterborne diseases [81], [82]. The 2022 flooding was devastating because the antecedent condition (e.g., high soil moisture) provided a favorable environment for the rainfall to translate into relatively higher runoff than other historical flood events. The rainfall was also record breaking across northeastern Bangladesh and the upstream of the basin (Cherrapunji and Shillong in India).

In this study, we first investigate the 2022 flood in northeastern Bangladesh to explore whether and how it was a compound flooding event, leveraging available hydrometeorological data.

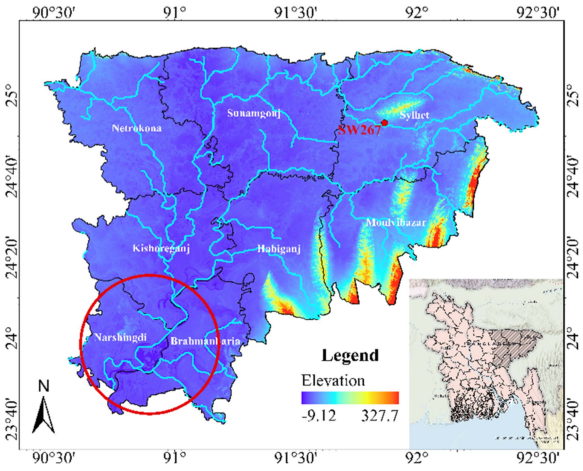

Fig. 1. Administrative boundary of Bangladesh and its northeastern region (inset), which is a part of the Megna River Basin. The red circle shows the basin outlet, and the red dot shows the stream gauge SW 267 in the Surma River near Sylhet (a transboundary river that drains upstream discharge into the region).

Then, we examine the potential for using SAR imagery to derive surface water (i.e., water extent climatology + flood) and flood extent maps for the 2022 flooding event by introducing a thresholding algorithm for flood detection. It is called the reference-based threshold method (RTM), where water extent climatology maps serve as references to dynamically determine the threshold for each target image for which floods are being detected. The threshold is then used to identify the watered pixels. Next, we evaluate the results by comparing them with those from other threshold-based approaches, local water-level data, and other global flood datasets. The innovation of this study is to develop a simple and rapid, yet efficient flood detection algorithm based on SAR images, which can be used to automate flood mapping while introducing an adjustment to capture the hydrological characteristics of the study area using monthly water extent climatology for each monsoon month, rather than a fixed permanent water body.

II. Data and Methods

A. Study Area

Bangladesh, a deltaic country located in South Asia, is one of the most flood-vulnerable countries in the world [83]. This study focuses on northeastern Bangladesh, which is a part of the Meghna River Basin and consists of eight districts (administrative units), as shown in Fig. 1. The geographic location of the region ranges between 23° 38' and 25° 12' North Latitude and 90° 27' to 92° 30' East Longitude, covering an area of about 21000 km^2. Northeastern Bangladesh has relatively low elevations, ranging from 4 to 9 m above mean sea level across most of the region, with some hilly areas as high as 20 m [84]. It experiences heavy rainfall during the monsoon season (May–October), which frequently causes flooding. Both pluvial and fluvial flooding events are evident in this region.

There are about 373 Haors in the study area, encompassing 859 km^2, 43% of the total Haors in Bangladesh [85]. Haors (locally named) are a type of wetland with bowl-shaped depressions filled with water over the monsoon and dry up during winter. During the monsoon season, localized rainfall and discharge from upstream fill up the dry Haors. The extent of water across the region varies monthly during the monsoon season [86]. Heavy rainfall and upstream discharge compromise the capacity of the Haors and cause flooding in the region [87], [88].

B. Data

1) *Satellite Data—Sentinel-1 and Landsat 8:* Sentinel-1 is a radar imaging mission conducted by the European Space Agency (ESA) that offers uninterrupted dual-polarization *C*-band SAR data with a temporal resolution of 6 or 12 days. Land masses across the globe are surveyed and mapped at repeat intervals of 12 days, while European lands are mapped within a 6-day repeat cycle [89]. In this study, the IW swath and the VV polarization are used because VV polarization provides better results in flood mapping than the other polarizations [60], [90]. This study also uses the United States Geological Survey's (USGS) Landsat 8 Level 2, Collection 2, and Tier 1 datasets. These datasets provide atmospherically corrected surface reflectance and land surface temperature derived from the data produced by the Landsat 8 OLI/TIRS sensors. The satellite images are collected and analyzed on the GEE platform. GEE provides a cloud-based computation platform that houses a comprehensive and continuously updated archive of remote sensing data of various scales that can freely be used for data analysis [1], [36].
2) *Hydrometeorological Data:* Soil moisture, precipitation, and river discharge data are used in this study to assess if the catastrophic flood of 2022 in the study area featured compound flooding characteristics. The soil moisture data are collected from NASAs Global Land Data Assimilation System datasets. Precipitation intensity data are collected from the integrated multisatellite retrievals for GPM's half-hourly precipitation dataset. We convert the half-hourly precipitation intensity to daily mean intensity (mm/h). River stage and discharge data are obtained from the Bangladesh Water Development Board (BWDB) and the GloFAS.
3) *Flood Data:* Since 2021, a near-real-time flooding monitoring component called GFM has been added to GloFAS [38]. Surface water extent maps (comprising flood and permanent water bodies) of the study area obtained from the GloFAS database are used for validation. The global flood database (GFD), available at 250-m resolution, is the first comprehensive satellite-based database for global flood risk [91]. It contains 18 years (2000–2018) of water extent and population exposure maps. The water extent maps are generated using the MODIS instrument onboard NASAs Terra and Aqua satellites.

C. Methods

1) Permanent Water Body Identification: The monthly climatology of water extent varies significantly during the monsoon months (May–October) due to the distinctive characteristics of the study area (i.e., Haors). Therefore, we define permanent

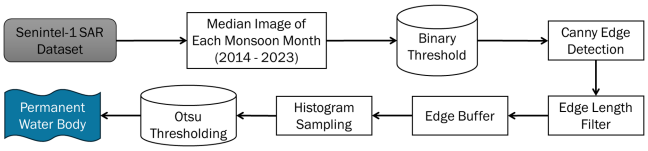

Fig. 2. Procedure for identifying and deriving permanent water body maps from SAR images using the edge Otsu method in the GEE environment.

water bodies as the monthly climatology of water extent and compute them for each monsoon month following the procedure, as illustrated in Fig. 2. However, if the water extent climatology does not vary monthly, the procedure will provide an identical permanent water body (water extent) map.

We employ the edge Otsu algorithm developed by Markert et al. [60] to delineate water, which executes multiple filtering and smoothing techniques to eliminate edge effects before applying the Otsu algorithm [92] for water delineation. However, unlike Markert et al. [60], we first generate the median of backscatter coefficient values at each pixel for each month using the respective Sentinel-1 images from 2014 to 2023. Then, a binary image is derived, defining whether the pixel's backscatter coefficient values (in decibels, dB) are less than an initial segmentation threshold value. A threshold of 16 dB is considered because Markert et al. [60] identified it as the optimum threshold through sensitivity analysis for Myanmar, which is close to our study area and shares similar geographical and climatic conditions. The initial threshold prevents the inclusion of edges from other classes, such as urban areas and forests, that may be present in the image. While we consider data from 2015 to 2023 due to data unavailability before 2015, the method can accommodate any period, depending on the data availability. Finally, the binary image is refined through multiple processes, such as Canny edge detection, edge length filter, edge buffer, and histogram sampling, before applying the Otsu algorithm to derive permanent water body maps for each monsoon month. The analysis is performed in the GEE environment.

2) Multisource Flood Mapping: We use the MSFM method proposed by Hamidi et al. [19], with a few modifications as required for the study area, to ensure a fair comparison to our method. A flow diagram of the method is presented in Fig. 1 of the Supplementary Material. The MSFM method uses optical (e.g., Landsat) and SAR (e.g., Sentinel-1) images for flood mapping. This approach involves multiple initial processing steps for postflood Landsat 8 optical images, such as generating mosaics, removing cloud shadows, and detecting water pixels using the modified normalized difference water index (MNDWI) in conjunction with Otsu's method for thresholding. The resultant optical flood map is subsequently used to calibrate the threshold for the normalized difference flood index (NDFI), which is derived from pre- and postflood Sentinel-1 SAR images. The optimal NDFI threshold is identified by systematically modifying the so-called k factor to enhance concordance between the NDFI and MNDWI maps. Its application may be limited to places with heavy clouds because optical images may not provide robust data on a cloudy day. Another potential limitation of the MSFM method is that it may not be suitable for automatically delineating flood extents simultaneously for multiple flood events, as pre- and postflood images must be identified diligently, which often requires manual investigation to determine the most suitable pre- and postflood images.

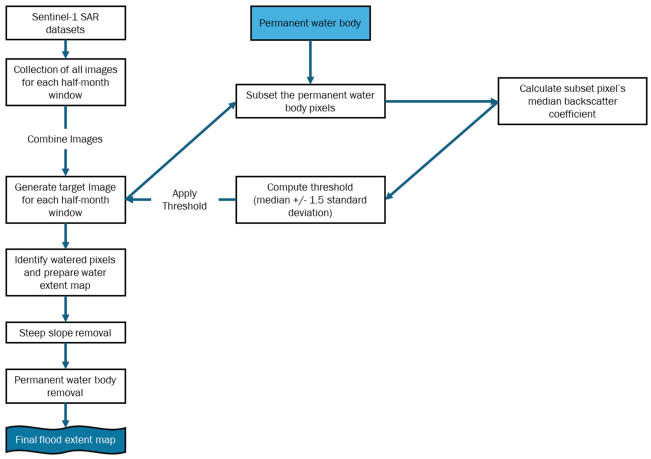

Fig. 3. Flow diagram of the adaptive threshold method for delineating flood extent using SAR images in the GEE environment.

3) RTM for Flood Mapping: We introduce an adaptive threshold method called the RTM for flood mapping. The method is simple and rapid, utilizing a reference permanent water body to dynamically determine the threshold that distinguishes between water and nonwater pixels. While the well-established threshold-based concept inspires our method, instead of using the backscatter coefficients from all pixels (watered and nonwatered), as is typically used in threshold-based algorithms, we utilized the backscatter coefficients from the permanent water body (i.e., water extent climatology in our case). This innovative modification enables us to eliminate erroneous flood detection by existing threshold-based methods, as computing the flood detection threshold considering all pixels (both watered and nonwatered) can be inaccurate, leading to excessive flood detection during minimal flood events. Moreover, instead of a single static permanent water body used in the typical SAR-thresholding algorithms, we considered separate permanent water bodies for each month (i.e., monthly climatology of water extent), which allows us to separate the seasonality in surface water extent in the study area, leading to the accurate estimation of floods (i.e., difference between surface water extent and the permanent water body). Unlike MSFM, RTM requires SAR images only and does not need to identify pre- and postflood images. In short, first, a threshold is determined from the backscatter coefficients of the pixels of the target image (for which water/flood extent will be computed) that correspond to the watered pixels of the permanent water body map. Then, the threshold is applied to the target image to identify the pixels that are watered. We use the method to derive surface water (permanent water + flood) and flood extent maps for the monsoon months (May and June) of 2022, when the catastrophic back-to-back flood occurred in northeastern Bangladesh (i.e., the study area). The flow diagram of the RTM is illustrated in Fig. 3. The following steps are performed for the RTM analysis in the GEE platform.

1) Collect all available Sentinel-1 images for the monsoon months (May–July) of 2022. The total number of images used in this study is listed in Table II of the Supplementary Material.
2) Generate the target images for each half month by combining all available Sentinel-1 images during the period (multiple Sentinel-1 images over a half-month time window are required to cover the study area). During overlapping multiple images, the minimum backscattering coefficients are considered to capture the maximum flood extent [93].
3) Mask the target image (generated in step 2) with the permanent water body (discussed and derived in Section II-C) of the corresponding month and subset the backscatter coefficients of the masked pixels of the target image. Calculate the median and standard deviation (SD) of the subset backscatter coefficients.
4) Compute the backscatter threshold for identifying watered pixels in the target image using the median and SD estimated in (3). Our thresholding method is inspired by the method followed by the articles [47] and [94], where we identify the threshold by analyzing the histogram of the backscatter coefficients of the masked pixels of the target image (step 3). We assume that the backscatter coefficients of water are the same or within a range for all Seltinel-1 images. Using a single threshold for identifying watered pixels can be highly sensitive and may lead to substantial biases in flood delineation. Therefore, to avoid biases and the influence of the extreme values of the backscatter coefficients of the masked pixels (which might be outliers or erroneous), we consider a range as the multitude of the SD from the median (i.e., median $\pm\, k * \mathrm{SD}$). We empirically analyze using many k values to identify the optimum value. We iteratively identify the watered pixels in the target image as the ones whose backscattering coefficients fall within a range of multitudes of the SD from the median (computed in step 3) for which the water extent (permanent water body + flood) map best matches the area with the darker pixels in the target image (in SAR image darker pixels represent the water). After analyzing numerous test cases, we found that "1.5" is the optimized k value; therefore, the ± 1.5 SD from the median (i.e., median ± 1.5 SD) is the optimal threshold range for deriving the water extent map in our study area. Therefore, the pixels with backscatter coefficient values that fall within the threshold range (spanning from median -1.5 SD to median +1.5 SD) are considered as watered pixels. However, it (the multitude of SD, i.e., 1.5) may differ in other areas. The median value is used instead of the mean because the mean could be influenced by outliers, such as the double-bounce effect caused by objects (e.g., boats) in the water, which may provide unusual backscatter coefficients. Generate the surface water extent (permanent water body + flood) map by labeling the watered and nonwatered pixels.
5) Pixels with steep slopes higher than 5° are removed from the surface water and flood maps, as floods are unlikely to occur there [19], [94]. The Shuttle Radar Topography Mission (SRTM) digital elevation data version 4 ("CGIAR/SRTM90_V4") dataset in GEE is used to identify steep slopes.
6) Derive the flood map by removing the permanent water bodies from the surface water extent map (derived in step 4).

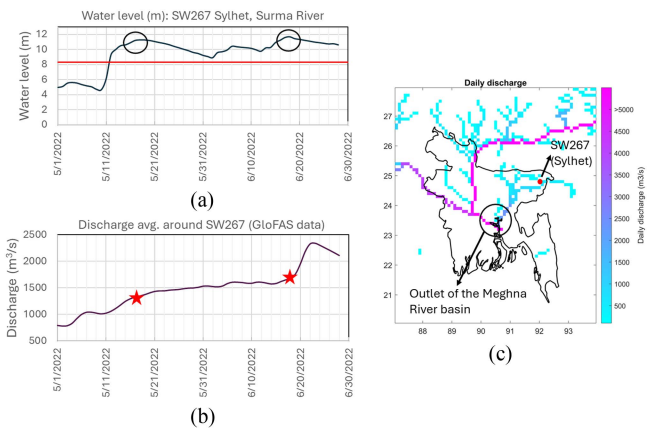

Fig. 4. (a) River stages (m) (i.e., water levels) and (b) discharges (m^3/s) time series for May and June 2022 at the station upstream of northeastern Bangladesh (SW267 Sylhet) located at Surma River near Sylhet City. The Surma River is one of the transboundary rivers that carries discharges from the upstream of the Megna River basin and drains through northeastern Bangladesh. The red line in the plot (a) represents the flooding threshold (i.e., danger level; 8.3 m) set by the Flood Forecasting and Warning Center of BWDB. (c) Daily discharge (m^3/s) in the major rivers across Bangladesh on 01 May 2022, obtained from the GloFAS datasets. The red point represents station SW267, and the black circle represents the approximate geographical location of the outlet of the Meghna River Basin, which receives flows from the northeastern region of Bangladesh and discharges to the sea (Bay of Bengal).

III. Results

In this section, we first analyze and discuss the compound nature of the 2022 flood event in northeastern Bangladesh, utilizing hydrometeorological data. We then illustrate the results from the MSFM method and evaluate its performance relative to the reference GFM data. Finally, the performance of the proposed algorithm (i.e., RTM) is compared with the other threshold-based flood detection algorithms [i.e., MSFM, the OTM, and global flood mapper (GFMapper)], local water-level data, and other global flood datasets (e.g., GFM) using different statistics, such as overall agreement, Kappa coefficient, and $F1$-score.

A. Analysis of the 2022 Flood in Northeastern Bangladesh

The 2022 flood consisted of two back-to-back flooding events in May and June 2022. Fig. 4 shows the time series of water levels and discharge at the stations (SW267) located near Sylhet city (upstream of the northeastern region of Bangladesh). The discharge series is derived by areal averaging the gridded GloFAS daily discharge upstream of the selected station (SW267) due to the unavailability of discharge data from BWDB. Plots show that the river stages (water levels) first exceeded the danger level, peaked around 18 May 2022, and then started receding, yet remained above the danger level before reaching a similar peak

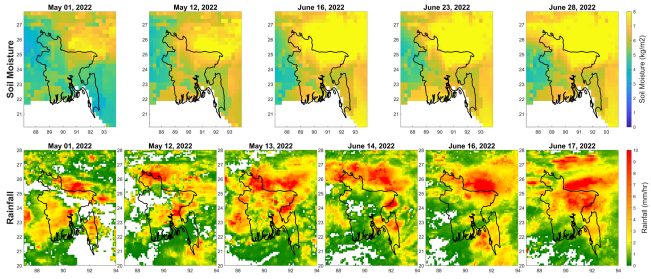

Fig. 5. Daily soil moisture and rainfall on different days of May and June 2022 during the 2022 flood events across northeastern Bangladesh.

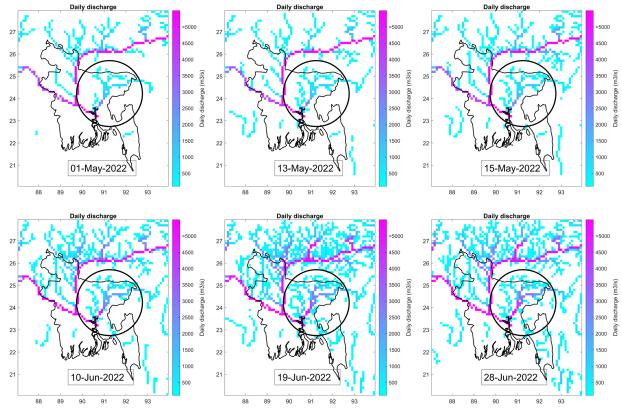

Fig. 6. River discharge on different days during May and June 2022. Graded color represents the discharge magnitude (m^3/s). The black circle highlights the changes in river discharge from the outlet of the Meghna River Basin to the upstream region (northeastern Bangladesh) on different days.

around 18 June 2022. This indicates two consecutive flooding events in the region. On the other hand, the discharge time series indicates that the discharge consistently increased over the entire period, even when the water level was reduced for a short period. The discharge continued to increase for a couple of days after the second peak in the water level. The consistent increase in discharge caused the second flood event to be even more devastating, as the river was already in bank-full conditions due to the first flood event. It started to spill over as discharge increased, which was not apparent in the water-level time series. This indicates that water level and discharge should be considered together for the accurate and efficient flood forecasting and monitoring.

The 2022 flood was driven by multiple hydrometeorological drivers forced simultaneously or in close succession. Those flood-generating drivers are as follows:

1) upstream discharge due to record-breaking rainfall across Assam, Cherrapunji, and Shillong (India) regions (upstream of the basin), which generates large discharge flowing toward the downstream northeastern region of Bangladesh;
2) heavy monsoon rainfall (convective and advective) across the northeastern region of Bangladesh (Sylhet and its surrounding areas);
3) antecedent wet condition (i.e., high soil moisture), which means that the basin was already sufficiently wet to generate more runoff and the downstream of the rivers (around the outlet of the basin) already absorbed massive discharge due to the first flood event (in May 2022) that occurred around a couple of weeks earlier than the second flood event occurred (in June 2022).

Fig. 5 shows the soil moisture and rainfall on different days in May and June 2022 when the floods occurred. The plot shows that the soil moisture on 01 May 2022 across the upstream Meghna River basin (Meghalaya, Assam, Cherrapunji, and Shillong in India) was relatively high. Hence, the rainfall across this area over the next few days caused relatively higher runoff. The extent and magnitude of soil moisture continue to increase over the remainder of May and June 2022, thereby elevating the magnitude of runoff and flooding. Fig. 5 also illustrates extreme rainfall in May and June 2022, which was one of the main drivers of the 2022 flood. Rainfall plots illustrate relatively low rainfall on 01 May 2022, while higher rainfall was observed on 13 and 14 May 2022, when the first flooding occurred. More widespread and intense rainfall was observed from 14 to 17 June 2022, during the second flood.

We utilize GloFAS daily discharge data to examine how discharge responded to heavy rainfall and high soil moisture during two flooding events in northeastern Bangladesh in May and June 2022. At the beginning of May 2022, the rivers across the northeast region and the Meghna River basin outlet had normal discharge (see Fig. 6). In mid-May 2022, discharge increased due to heavy rainfall, triggering the first flooding event. Despite no heavy rainfall for the next 2–3 weeks, the soil was highly saturated and even continued increasing across the region due to continuing average rainfall (soil moisture on 16 June 2022 compared with 12 May 2022, in Fig. 5). At the same time, river flows from the northeastern region accumulated downstream, causing relatively higher discharge at the basin outlet (discharge maps on 13 and 15 May and 10 June 2022, in Fig. 6).

The second flood occurred in mid-June 2022, when the region experienced heavy rainfall again, which began around 14 June 2022 and lasted for a few days. Due to the first flood event (in mid-May 2022), the soil was already highly saturated, and the outlet of the basin had already received numerous discharges from upstream. Therefore, during the second flood event (June 2022), it was unable to adequately drain additional discharge from the upstream, resulting in a backwater effect. The heavy rainfall in mid-June produced significant runoff, and the rivers spilled over due to the high discharge (discharge maps on 10 and 19 June 2022, in Fig. 6), causing the mid-June flood to be even more devastating and widespread than it might have been if there had been no earlier flood event. This demonstrates the compound flooding phenomena. There was sustained high discharge across the region over the next few days following the second flood event (discharge map on 28 June 2022, in Fig. 6), resulting in widespread waterlogging that persisted for several months.

B. *Permanent Water Body Identification*

We define the permanent water body as the water extent climatology (average over a decade/more extended period) across

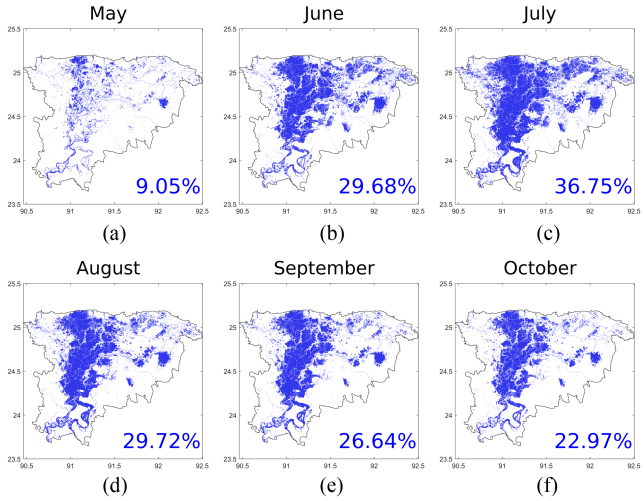

Fig. 7. Extent of the permanent water body (pixels with blue color) for (a) May, (b) June, (c) July, (d) August, (e) September, and (f) October. The white color depicts no water, and the black line represents the administrative boundary of the study area. The numeric values in the subplots represent the percentage of the study area covered by the permanent water body for different months.

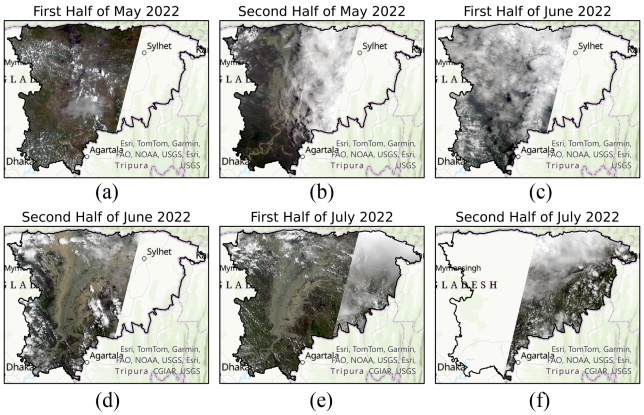

Fig. 8. Landsat-8 image for the first half and second half of (a) and (b) May, (c) and (d) June, and (e) and (f) July 2022.

the region. As discussed earlier, the permanent water body maps are expected to vary by monsoon month (May–October). Fig. 7 illustrates the permanent water body maps for each monsoon month. The plots show that the extents of the permanent water body vary depending on the monsoon months. At the start of the monsoon season (in May), the extent of the permanent water body is minimal (9.05% of the study area) compared with the other months. As the monsoon progresses, the extent of the permanent water body reaches its maximum in July (36.75% of the study area) and then decreases in the subsequent months.

C. Results From the MSFM Method

First, we investigate how suitable the Landsat-8 images are for mapping the flood during May and June 2022 across northeastern Bangladesh. Fig. 8 illustrates the mosaic of the least clouded Landsat-8 images for the two halves of the selected monsoon months, May, June, and July in 2022. It reveals that

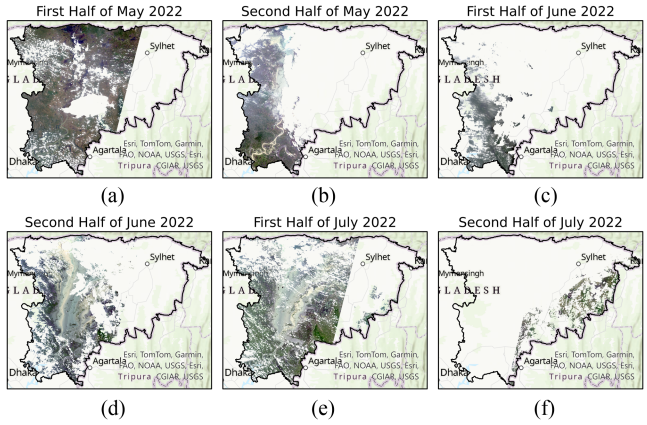

Fig. 9. Landsat-8 images after cloud and cloud shadow removal for the first half and second half of (a) and (b) May, (c) and (d) June, and (e) and (f) July 2022.

the Landsat-8 images suffer from significant cloud cover and do not cover the entire study area in most cases.

After removing clouds and cloud shadows, the data remaining in the Landsat-8 images (i.e., the area where image data are available) are significantly limited for generating flood maps, as illustrated in Fig. 9. After removing clouds, the image of the first half of May 2022 provides the maximum coverage of the study area compared with the other images, yet almost the entire eastern portion of the study area is missing, while the other images provide significantly lesser coverage of the study area, indicating that neither image is suitable for mapping the 2022 flood in northeastern Bangladesh. The Landsat-8 images, as shown in Fig. 9, do not cover the entire study area and are affected by cloud cover; therefore, flood mapping using these images is limited.

As the available Landsat-8 images are unsuitable for mapping the 2022 flood in northeastern Bangladesh, the SAR images from the Sentinel-1 satellite are used along with the Landsat-8 images to derive flood maps using the MSFM algorithm. We combined the MSFM-derived flood maps with the permanent water body maps of the corresponding month to derive surface water extent maps. Fig. 10 illustrates the surface water extent maps derived by the MSFM [see Fig. 10(e)–(h)] algorithm and the corresponding maps provided by GFM (reference maps) [see Fig. 10(a)–(d)] for the first and second half of May and June 2022.

The watered pixels in the surface water extent maps from MSFM (and GFM) are 15.45% (14.93%) and 41.44% (33.62%) for the first and second half of May 2022, respectively; they are 60.66% (37.80%) and 56.95% (51.48%) for the first and second half of June 2022, respectively. While the MSFM method reasonably captures the surface water extents, it significantly differs from the GFM water extent map for the first half of June 2022. This is also evident in Table I, where the efficiency statistics for the MSFM in the first half of June are lower than the others; specifically, the precision falls to 0.57, indicating that many pixels are incorrectly identified as flooded compared with the reference GFM data. This is because the Landsat-8 image for the first half of June 2022 covers a small portion of the study

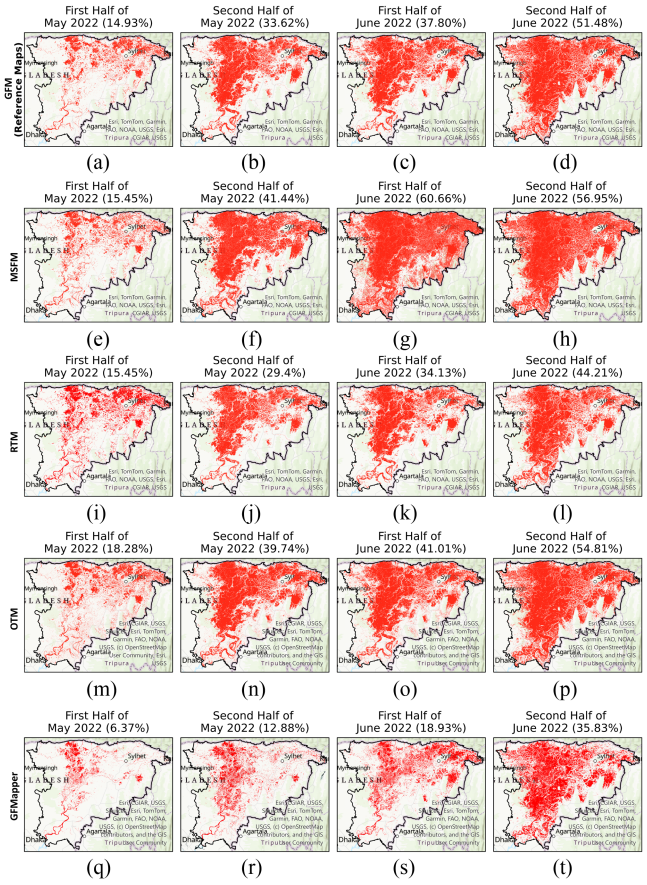

Fig. 10. Surface water extent during the (a), (e), (i), (m), and (q) first and the (b), (f), (j), (n), and (r) second half of May 2022 and the (c), (g), (k), (o), and (s) first and (d), (h), (l), (p), and (t) second half of June 2022 from the MSFM, GFM, RTM, OTM, and GFMapper. The maps from the GFM are the reference maps (a, b, c, and d). The numeric values in the titles of the subplots, within parentheses, represent the percentage of inundation of the study area.

TABLE I
EFFICIENCY OF THE SELECTED THRESHOLD-BASED FLOOD MAPPING METHODS, INCLUDING THE PROPOSED RTM METHOD IN COMPARISON WITH THE GFM (REFERENCE) DATA

Month	Half	Method	Overall Accuracy	Kappa Coefficient	Precision	$F1$-score
May	First	RTM	0.96	0.95	0.74	0.76
May	First	MSFM	0.95	0.94	0.73	0.74
May	First	OTM	0.97	0.97	0.84	0.85
May	First	GFMapper	0.92	0.92	0.97	0.15
May	Second	RTM	0.95	0.93	0.92	0.86
May	Second	MSFM	0.92	0.87	0.74	0.82
May	Second	OTM	0.96	0.94	0.85	0.90
May	Second	GFMapper	0.86	0.83	0.85	0.47
June	First	RTM	0.90	0.90	0.96	0.86
June	First	MSFM	0.84	0.70	0.58	0.72
June	First	OTM	0.97	0.95	0.92	0.93
June	First	GFMapper	0.87	0.84	0.92	0.62
June	Second	RTM	0.92	0.87	0.96	0.85
June	Second	MSFM	0.92	0.86	0.84	0.85
June	Second	OTM	0.96	0.94	0.93	0.94
June	Second	GFMapper	0.89	0.84	0.96	0.79

area (with limited data) when clouds are removed [see Fig. 9(c)], which generates errors in identifying the NDFI threshold used for deriving flood maps from the Sentinel-1 image, leading to the overestimation of the flood extent. It is worth noting that the NDFI threshold is derived by iteratively comparing the NDFI-based flood map (derived from the Sentinel-1 image) with the MNDWI-based flood map (derived from the Landsat-8 image). Therefore, using the Landsat-8 images with significant cloud cover may cause errors in flood maps derived using the MSFM method.

D. Results From the RTM

The RTM provides comparable flood detection with the MSFM method while overcoming the limitation of detecting floods in the presence of clouds. Results show that approximately 60.66% of the pixels are watered when the MSFM method is applied, compared with the 37.8% of watered pixels found in the GFM reference map during the first half of June 2022. In contrast, RTM estimates that 34.13% of the pixels are watered. Besides, RTM estimates that 15.45% , 29.4% , and 44.21% of the pixels are watered for the first half of May 2022, the second half of May 2022, and the second half of June 2022, respectively, which are comparable with those found in GFM, as illustrated in Fig. 10.

The flood detection efficiency of RTM is also comparable with that of the MSFM, as illustrated by the comparable values of accuracy and Kappa coefficient (except the first half of June 2022) (see Table I). However, the higher precision of RTM compared with MSFM (except the first half of May) suggests that RTM can better classify the flooded areas without misclassifying nonflooded areas as flooded. In the first half of June, even though both methods have the overall high accuracy, the Kappa coefficient is much lower for MSFM (0.7) compared with RTM (0.9), indicating that RTMs flood detection is much more reliable and less influenced by chance [95]. The $F1$-score is comparable for both methods, with RTM slightly favoring.

We further validate the proposed method's performance by comparing its accuracy with the existing SAR-based flood mapping methods, such as the OTM [52] and the GFMapper [96]. Fig. 10 shows the maps generated by OTM (m–p) and GFMapper (q–t). It illustrates that OTM achieves a similar water extent (a 4% –5% difference) to GFM for all halves of May and June 2022. On the other hand, GFMapper significantly underestimates the water extent compared with that obtained from GFM. Similar conclusions are also evident in the efficiency statistics, as tabulated in Table I. GFMapper exhibits low efficiency, especially during the first half of May, when flooding is minimal, suggesting that the model may not perform well in relatively dry conditions. This phenomenon is also observed for the OTM, as shown in Fig. 11, illustrating the water extent delineated using (a and d) GFM, (b and e) RTM, and (c and f) OTM for the first half of May 2020 and 2021, respectively. The results show that the OTM overestimates the water extent (35.83% and 28.31%), whereas the RTM provides water extents (9.42% and 10.10%) comparable with those in the reference GFM maps (6.31% and 6.33%). This limitation of the OTM may be due to the

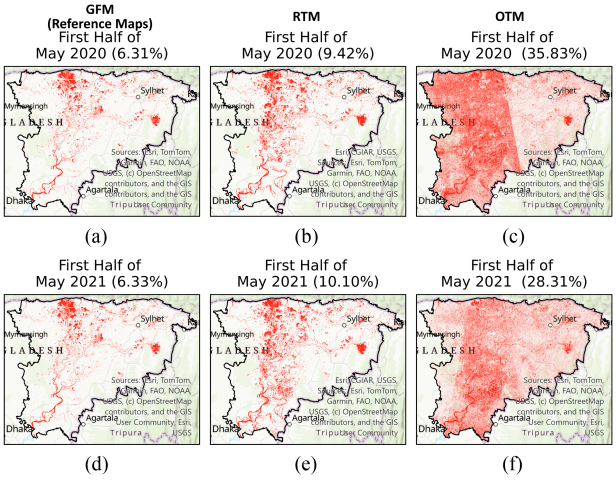

Fig. 11. Surface water extent during the first half of May 2020 and 2021 from (a) and (d) GFM, (b) and (e) RTM, and (c) and (f) OTM. The numeric values in the titles of the subplots within parentheses represent the percentage of inundation of the study area. The maps from the GFM are the reference maps (a and d).

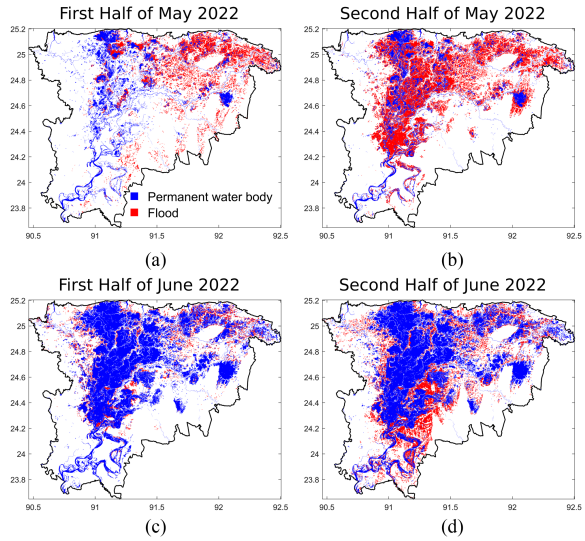

Fig. 12. Flood extent maps along with permanent water bodies for the 2022 flood in northeastern Bangladesh for the first and second half of (a) and (b) May and (c) and (d) June derived by the adaptive threshold flood mapping method. The blue pixels represent the permanent water for the specific months (which vary with the monsoon months), and the red pixels denote the flooded pixels.

water delineating threshold being computed based on all pixels (watered and nonwatered) in the images, resulting in an erroneous threshold. In contrast, the threshold in RTM is computed using the backscatter coefficients of pixels corresponding to the watered pixels in the water extent climatology map, leading to a robust threshold.

To further validate the RTMs flood detection ability for the 2022 flood, we use the river stages at 13 stations across the study area to investigate if our surface water extent maps identified the area (pixels) around the stations that are flooded when the stages at the respective stations exceed the danger level (i.e., flooding threshold) set by the BWDB. It is worth mentioning that high watermarks and aerial imagery data are unavailable for the 2022 flood event. Out of 26 instances (back-to-back floods in the second half of May and June $\times$ 13 stations), 5 were incorrectly identified by the RTM (3 were incorrectly identified as flooded, and 2 were misclassified as not flooded), leading to 81% accuracy (Fig. 3 of the Supplementary Material). Furthermore, RTMs surface water extent maps are compared with those provided by the GFD for two flood events available in the study area (25 July 2016 and 10 August 2017), as illustrated in Fig. 2 of the Supplementary Material. To eliminate the mismatch in the resolution of the RTM and GFD flood maps, we used ArcGISs nearest neighbor resampling method, which assigns a cell to be flooded or not based on the pixels located at the center of each 250 m cell of the RTM flood maps. For both events, the surface water extent maps derived from the RTM are comparable with those of GFD. For the July 2016 event, the estimated water extents are 37.34% and 40.53% of the study area computed by the GFD and the RTM, respectively. According to the GFD, the surface water extent for the August 2017 event is 36.35% , which is 35.55% when the RTM is applied. This indicates that the results from the RTM are comparable with those from GFD.

Finally, we employ the RTM to derive the flood extent maps for the back-to-back floods in May and June 2022 across the northeastern region of Bangladesh, which is illustrated in Fig. 12. Results indicate that, during the first half of May 2022, the flood started and flooded the far northeastern region of the study area (i.e., Sylhet), where the transboundary rivers (e.g., Surma and Kushiara) bring discharge from the upstream basin [see Fig. 12(a)]. Over time, the flood spread across the region and continued to propagate downstream during the second half of May 2022. During June, the extent of the permanent water body (climatology of water extent) is more extensive than that of May. Therefore, many flooded (red) pixels in the second half of May 2022 [see Fig. 12(b)] are now a part of permanent water bodies for the first and second halves of June 2022 [see Fig. 12(c) and (d)]. However, many new pixels flooded during June 2022 compared with May 2022. Results show that the flooding over the second half of June is more extensive than the first half, when the second flood struck. The surface water extent maps (see Fig. 12) indicate the widespread inundation across the study area from the back-to-back 2022 floods in northeastern Bangladesh.

TABLE II
PERFORMANCE OF THE RTM IN TWO FLOODS IN MAY 2020 AND OCTOBER 2023 IN NORTHEASTERN BANGLADESH FOR FURTHER VALIDATION USING GFM (REFERENCE) MAPS

Flood	Overall Accuracy	Kappa Coefficient	Precision	*F*1-score
May, 2020	0.96	0.95	0.86	0.81
October, 2023	0.96	0.95	0.93	0.87

After evaluating the method's performance for the 2022 compound flood, we assessed its performance in deriving water extents during other periods and for various scales of inundation. To achieve this, we mapped the water extents caused by low and moderate floods in May 2020 and October 2023, respectively, as shown in Fig. 13. Overall, RTM provides nearly the same water extents as the reference GFM maps. The overall accuracy of 0.96 and a Kappa coefficient of 0.95 for both cases (see Table II)

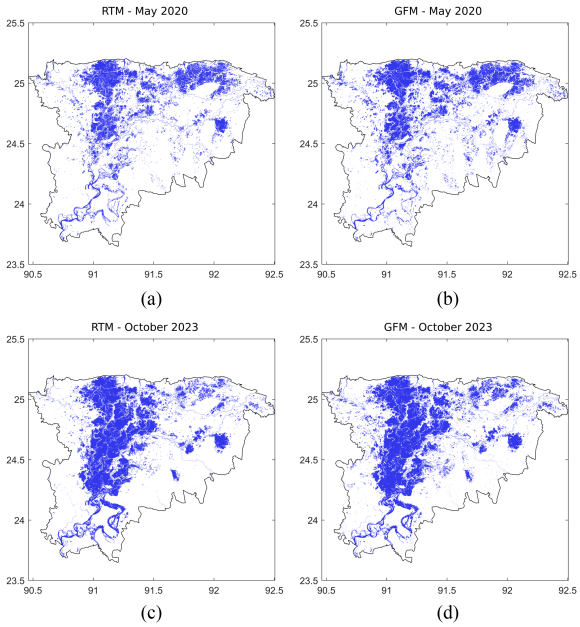

Fig. 13. Comparison of water extent maps between (a) and (c) ATM and (b) and (d) GloFAS for the floods in May 2020 and October 2023, respectively.

indicate the robustness of the proposed method in mapping water extent in northeastern Bangladesh.

IV. Discussion and Conclusion

The study formulates a simplified yet efficient method called the RTM, which is an adaptive algorithm for mapping surface water (water extent climatology + flood) and flood extents utilizing SAR data, employing dynamic hydrological references instead of static global thresholds. RTM utilizes past water extent climatology (monthly "permanent water" maps) as a context-specific mask and calculates a robust threshold from the backscatter of known-water pixels. The use of climatological water bodies enables the consideration of documented seasonal variation in wetlands, such as the Haors in Bangladesh, which experience annual flooding and drying cycles. The innovation avoids the misclassification of typical seasonal water as flood and vice-versa, a feature absent in previous SAR-thresholding algorithms. This approach is entirely automated in GEE, eliminating the need for human pre- or postevent picture matching or optical data, as required in MSFM [19], and seamlessly adjusts to seasonal and regional hydrological fluctuations. Unlike the traditional histogram-based techniques (e.g., Otsu), which utilize a singular static threshold that may inaccurately categorize water due to overlapping backscatter distributions, RTMs median-based dynamic thresholding provides a more resilient and adaptable approach to flood delineation. We demonstrate how the most recently developed satellite image-based flood mapping algorithms, such as MSFM [19], can be limited for our study area (where floods often occur during monsoon months when extensive clouds are present) due to the significant presence of clouds in the satellite optical images, even after the clouds are removed. In contrast, our method provides reliable and robust surface water and flood extent across the area, comparable with the other existing SAR-based models and even better in a few instances, particularly for low-flood events. The method does not require preflood images (a common requirement in most existing methods, e.g., MSFM) to delineate flood extent. It allows the automatic application of as many flood events as needed. Estimation of the monthly varying permanent water extent is a crucial modification added to the algorithm that addresses the unique hydrological characteristics of the study area (i.e., Haors), where the climatology of permanent water bodies varies with monsoon months while also applicable to other places where they do not differ much with months. The integration of SAR-only data, monthly climatological references, and automated processing yields a rapid, reproducible workflow applicable to many geographies and flood types.

We apply the proposed reference-based thresholding flood mapping method (i.e., RTM) to delineate the surface water and flood extents for the back-to-back floods in northeastern Bangladesh that occurred in May and June 2022. Using the hydrometeorological data, we identify that the 2022 flood is a perfect example of a compound flood. It first occurred in mid-May 2022, driven by the local heavy precipitation and upstream discharge, while a precondition of high soil moisture provided a favorable condition for exacerbating the flood impacts. Later, a subsequent flood event occurred in mid-June 2022, before the effects of the first flood had been recovered, making the second flood even more widespread and destructive.

The study demonstrates the capability of SAR imagery and reference-based thresholding flood mapping methods in estimating surface water and flood extent for the 2022 compound flood in northeastern Bangladesh. The rapid flood detection and mapping method developed in this study is particularly crucial for resource- and data-limited regions and geographies that experience monsoon floods with frequent cloud cover. The scripts associated with the method will be publicly available and can be run in the GEE environment to derive flood maps for any area of interest. They would be helpful for flood risk assessments and mitigation, as well as support emergency responders and decision makers with valuable flood information for resource allocation and planning rescue operations. On the other hand, the GFM flood maps require approximately 8 h after the satellite images are publicly available due to their global coverage and complex algorithms. Decision makers will not need to wait until the GFM flood maps are available; instead, local experts with minimal training can use our method to derive rapid and robust flood maps within the GEE environment, saving precious time amid flooding. While the water extent maps available on GFMs web portal rely on Sentinel-1, the proposed method allows users to easily use any SAR images from any satellite for flood mapping, enabling the reconstruction of flood maps for historical periods.

This study utilizes monthly varying water extent climatology maps for the monsoon months, rather than a single, fixed, and permanent water dataset, because the study area exhibits substantial monthly water extent variations during the monsoon months. This allows robust flood delineation by subtracting the water extent climatology from the surface water extent (i.e., water extent climatology + flood). Although it is outside

the scope, future studies may consider investigating dynamic changes in water resources and floods during the monsoon months over time, using a single permanent water extent (i.e., the lowest water extent during the nonmonsoon months). While Sentinel-1 images are used to derive the water extent climatology, future studies could incorporate historical optical satellite images and their products, such as the USGSs Dynamic Surface Water Extent (DSWE) [96]. However, a quick analysis reveals that they are unsuitable for studying the 2022 flood in northeastern Bangladesh due to significant cloud cover (see Fig. 4 of the Supplementary Material). Notably, DSWE is derived from Landsat (optical) images. Moreover, the fusion of various satellite sources may reduce reliance on a single satellite source, as evident in the failure of Sentinel-1B and the decommissioning of other satellites [97], [98]. Although SAR pictures provide a limited temporal record, optical images may function as a viable alternative for reconstructing flood maps during extended historical periods. Nonetheless, clouds and shadows in optical pictures continue to limit their use in the creation of flood maps for extensive historical durations. Recent research has devised techniques, such as deep learning, to deduce inundation maps for an extended historical time by utilizing the correlations between SAR and accessible cloud-free optical pictures [99]. In addition, available methods for removing building shadows from optical images can make the images suitable for flood mapping in highly urbanized areas [100], [101], [102].

The proposed flood mapping method (RTM) is scalable and transferable; we tested the proposed method to map the water extent for Hurricane Harvey (2017) across Galveston Bay, TX, which reveals good performance (see Table 2 and Figs. 5 and 6 of the Supplementary Material). The results showed better agreement with an $F1$-score of 0.96 when using the "median $\pm$ 1.75 SD" threshold, compared with 0.87 with "median $\pm$ 1.5 SD." This indicates that the flood detection threshold may need to be adjusted and optimized for new study areas. While the "median $\pm$ 1.5 SD" was the most suitable for detecting floods in our study area, a different multiplier of SD might be appropriate for other study areas [19], [103]. An automatically selected optimum threshold, determined by comparing surface water extent maps to reference maps in terms of percentage area inundated and efficiency statistics, such as the Kappa coefficient and precision, should enable the RTM method to be efficiently applied to other regions. In addition, one can optimize model efficiency by simultaneously tuning the threshold (i.e., the multiplier of SD) and investigating whether the water extent maps show flooding around the river stations where observed river stages exceed the flooding thresholds when no high-water mark data are available (as illustrated in Fig. 3 of the Supplementary Material). Otherwise, high-water mark data can be used for this analysis. Therefore, the proposed RTM method can be applied to any region for flood extent mapping by identifying an optimized threshold. One of the advantages of using satellite imagery in flood mapping is that it can capture all types of floods, including fluvial, pluvial, and their combination (i.e., so-called compound flood). While the study focused on the extreme flood in 2022 across northeastern Bangladesh, we also investigated the water extent maps for other low- to moderate-flood events that occurred in May 2020 and October 2023, revealing the fidelity of the RTM method for flood mapping. However, there is still scope for extending this work to investigate whether and to what extent the method may exhibit overdetection of floods when applied to heterogeneous flood-prone regions.

The proposed method enables rapid mapping of floods on a publicly available cloud-based computation platform (i.e., GEE) with minimal resources and expertise. It should be a helpful tool for data-scarce and resource-limited regions, enhancing flood resilience and the capacity of local administrations by providing valuable flood information for decision makers and first responders. Inspired by the successful mapping of floods, we envision potential future research to develop a half-monthly time series of water extent maps, thereby deriving a probabilistic inundation map that quantifies the annual flooding probability at each pixel across the study area. Another potential future extension of this study is to integrate depth estimation algorithms with datasets, such as population, agriculture, and built environment to assess the impacts of floods on society and the environment.

IV. Data Availability

Sentinel-1 comprises public domain datasets offered by the ESA, accessible via the Copernicus Data and Information Access Service portal at https://sentinel.esa.int/web/sentinel/sentinel-data-access. Landsat imagery is public domain data provided by the USGS under the National Land Imaging program: https://www.usgs.gov/Landsat-missions. Sentinel-1 and Landsat datasets are accessible via the Google Earth Engine data catalog: https://developers.google.com/earth-engine/datasets.

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